Taiwo Alabi

Backend Engineer (Python), AI Engineering: Agent Foundations

newalabitaiwo625@gmail.com | +353 87 004 2838 | linkedin.com/in/tami-alabi | github.com/iamtamilore | iamtamilore.github.io | County Dublin D03 K0Y3, Ireland

Profile

I build Python backend services that run AI agents in production, and I write the evaluations that prove the agents behave. I built the multi-model gateway that every model call on my machine passes through, with a response cache, a per-call budget guard, a daily spend ceiling and a detector that quarantines reasoning burns. On 10 November 2026, its ledger held 33,104 calls with token counts, endpoint and cost. Around it I wired eight MCP servers into one agent layer, added a watchdog that restores a dead server with no human intervention, and built the scheduling layer for 18 production jobs so a job that fires and fails is distinguishable from a job that never fires. The Smart Medical Records Engine is a clinical assistant on FastAPI and PostgreSQL with 768 dimensional vector search, running across four Docker Compose services with retrieval scoped per patient, physician authorization, PII redaction and an audit log of every access. The Customer Care Process Automation system is an AI routing agent reaching 95.2% routing accuracy on 42 held out tickets, behind a confidence gate that escalates uncertain tickets to a human instead of guessing. At Huawei I automated CRM extraction in Python, validated 100k+ rows of engagement data in SQL, and helped design over 440 User Acceptance Test cases for a data migration covering 400 million daily call records. MSc Artificial Intelligence, National College of Ireland, completing January 2027, and available for full time work from that date. Based in Dublin, working remotely across Irish and UK timezones.

Selected Systems

Multi-model LLM gateway: Python, sqlite, kie.ai, OpenRouter, NVIDIA NIM

- Problem: Every model call was a direct provider call, so there was no cache, no per-call budget, no spend ceiling, no lineage, and no way to tell a reasoning burn from a real answer.
- Action: I built one chokepoint for every call, with a sha256 response cache keyed on model, messages and parameters, a per-call budget guard, a daily cumulative ceiling read from the ledger, and a detector that quarantines a response where completion tokens equal reasoning tokens and content is empty.
- Result: The ledger held 33,104 calls on 10 November 2026, with token counts, endpoint and cost. The registry separates vendor facts, which are fetched and overwritten, from user facts, which are hand-kept and never overwritten by a fetch, so a scheduled catalogue sync cannot silently drop a routing decision.

Multi-agent MCP layer: MCP protocol, Python, Node.js, opencode, Claude Code

- Problem: Each AI client on the machine had its own route to models, memory, code, the web and the browser, so every client drifted into different behaviour.
- Action: I wired eight MCP servers into one layer, unified-log, kie-ai, code-search, claude-mem, ask, comfyui, wikipedia and browser, with six also in the global config so a chat outside the project gets the same core set. The model gateway is exposed as MCP tools, so an agent inside a chat reaches models through the same budget, cache and ledger as the command line.
- Result: A watchdog tails the transport error log, bumps the affected server’s restart marker in both config files, and the client respawns the server, verified end to end on a SIGTERM with no human intervention.

Cron orchestration with a verification ledger: bash, Python, JSONL ledger, systemd

- Problem: 18 scheduled jobs were failing silently. A job that fires and fails and a job that never fires leave identical evidence in a log file.
- Action: I routed every crontab line through one runner with a job name argument and made the orchestrator compare the job registry against an append-only ledger, writing an explicit per-run outcome so the two failure modes are distinguishable. The orchestrator self-heals the one repairable drift, and notifications fire only on a state change, because a monitor that reports a known-bad state every fifteen minutes trains the reader to ignore it.
- Result: 18 jobs under one registry and 3,284 ledger records, with the exit code visible to anything watching it, 0 healthy and 1 not.

Smart Medical Records Engine: FastAPI, PostgreSQL with pgvector, Docker Compose

- Problem: A doctor has ten minutes and a patient history spread across years of notes, and the one critical line, a past reaction to a medicine, gets missed.
- Action: I built a clinical assistant that answers plain English questions from that patient’s own records, with retrieval scoped per patient so no other patient’s notes can surface, plus physician authorization, PII redaction, and an audit log of every access. Thirty adversarial probes tried to pull another patient’s note and none got through.
- Result: I ran the whole stack, the API, PostgreSQL with 768 dimensional vector search and a self hosted Llama 3.1, across four Docker Compose services, so patient data never leaves the hospital’s own network.

Evaluation and Quality Engineering

Turing, Remote (Dublin): Apr 2026 to May 2026 **AI Quality Analyst, Google Gemini evaluation programme**

- Measured the quality of Google Gemini responses on personalization features.

- Wrote prompts to test the models and documented the process for a Reinforcement Learning from Human Feedback project.

Automated QA Gate for AI Generated Imagery: TensorFlow/Keras, ResNet50, scikit-learn

- Built a gate over frozen ResNet50 Siamese features that reaches precision 1.00 at F1 0.90 on 90 sealed test pairs, against pixel difference and kNN baselines at F1 0.59 and 0.26.
- Tuning the decision threshold from the sigmoid default of 0.5 to 0.4821, on validation data only, recovered all eight missed pairs without adding a single false positive and lifted F1 from 0.90 to 1.00.
- Public write up with ROC curve and confusion matrix. Read the write-up (PDF)

Professional Experience

Huawei Technologies (via RGS), Globacom Telecommunications department, Lagos

Data Analyst Intern: Mar 2025 to Jul 2025

- Worked with stakeholders to identify data and reporting requirements, turning raw data into relevant and meaningful insights.
- Managed Power BI assets, including reports, dashboards, workspaces, and underlying datasets, and designed scalable data models with advanced analytics enabled in the reporting.

Data Specialist: Jul 2025 to Dec 2025 (internship extended by the team)

- Created database structures and tables from 20+ business requirements, and dashboards in SSRS and Tableau showing 30+ KPIs across departmental operations.
- Wrote Python scripts to automate daily CRM data extraction, cutting manual reporting time by 15 hours per week.
- Ran SQL queries to clean and validate 100k+ rows of user engagement data, which surfaced a 14% discrepancy in monthly active user metrics.
- Worked with the solutions architect on a data migration architecture for 400 million daily call records, partitioning customer data into domain specific buckets, and with the test team designed over 440 User Acceptance Test cases across Q2 to Q4 so faults were caught before anything went live.

Education

National College of Ireland, Dublin: Jan 2026 to Jan 2027 **MSc Artificial Intelligence** Modules: Machine Learning, Intelligent Agents and Process Automation, AI Driven Decision Making, Engineering and Evaluating AI, Natural Language Processing, Data Governance and Ethics, Database Programming, Foundations of AI.

Afe Babalola University, Lagos: Sep 2021 to Feb 2025 **BSc Computer Science, Second Class Honours, Upper Division (2.1)**

Technical Skills

Python backend and APIs: Python, FastAPI, Flask, REST services, asynchronous and background processing patterns, data models, PostgreSQL, SQL, data modeling, SQLite.

Agent platforms: MCP servers, agent orchestration, multi agent flows, tool calling, model routing, fallback chains, LLM gateways, RAG pipelines, LLM evaluation, vector search (pgvector, FAISS), Claude API, LangGraph, confidence gating and human escalation.

Testing and operations: pytest patterns, automated evaluation harnesses, confusion matrix analysis, ROC curves, cron and scheduler orchestration, append-only ledgers, self-healing watchdogs, Docker, Docker Compose, Kubernetes, Google Cloud, autoscaling.

Data and governance: Power BI, SSRS, Tableau, GDPR data cataloging, data lineage, data quality validation, UAT design.

Tools: Git, GitHub, Claude Code, native Linux, Microsoft 365.

Certifications

Kubernetes: clusters, pods, auto scaling, memory management, distributed systems (Udemy, 2026): View certificate

Work Authorisation

Irish Stamp 2 during my MSc: 20 hours per week in term and 40 hours per week over the college holidays, including December. I complete my MSc in Artificial Intelligence in January 2027 and am eligible for Irish Stamp 1G post-study graduate work permission from that date, which permits full time employment at 40 hours per week. Based in Dublin, Ireland, working remotely in Irish and UK timezones.